

Personal Information

Seyed Soroush Pakzad

- ✓ Civil & Structural Engineer
- ✓ Email: soroush.pakzad@gmail.com
- ✓ Phone: +64 22 458 0435
- ✓ Address: Euston Street, Riccarton, Christchurch, 8041, New Zealand
- ✓ LinkedIn Account: [Soroush Pakzad](#)

Professional Profile

Civil and Structural Engineer with over two years of professional experience in technical office engineering, project planning and coordination, construction supervision, engineering documentation, quantity surveying, and cost control. Experienced in project scheduling, progress monitoring, technical coordination, and multidisciplinary project delivery.

Research background in structural engineering, with a focus on Structural Health Monitoring, bridge damage detection, machine learning, and advanced concrete materials. Experienced in vibration-based structural assessment, anomaly detection, and data-driven analysis. Proficient in AutoCAD, SAP2000, ETABS, SAFE, ABAQUS, Python, and MATLAB, with an analytical and detail-oriented approach to solving engineering problems.

Work Experiences

Technical Office Engineer | Gostareshe Rah & Abniye Shargh Consulting Engineers Company,
Mashhad, Iran (Website: gostareshrah.com), Full-Time

May 2024 - Aug 2026

- **Project Planning & Management**
 - Project planning, scheduling, progress monitoring, and preparation of project reports
 - Coordination of multidisciplinary engineering and construction activities
 - Supporting project management and procurement processes
- **Quantity Surveying & Cost Control**
 - Quantity surveying and measurement of construction works
 - Monitoring project costs and supporting cost-control activities
 - Review of quantities, contracts, and technical documentation
- **Construction & Technical Coordination**
 - Technical coordination between engineering teams, contractors, and project stakeholders
 - Construction supervision and monitoring of compliance with technical requirements
 - Identification and resolution of technical issues during construction
- **Selected Projects**
 - Dormitory for Married Students, Ferdowsi University of Mashhad
 - Termeh Commercial-Office Building Project (Procurement & Construction)

Researcher | Department of Civil Engineering, Ferdowsi University of Mashhad (FUM),
Mashhad, Iran (Website: um.ac.ir), Remote

Sep 2023–Aug 2026

Selected Projects:

- **Early Damage Detection in Bridges Using Hybrid Unsupervised Machine Learning Frameworks**
 - Developed and evaluated VAE-based hybrid unsupervised learning frameworks for early bridge damage detection using real-world Z24 Bridge vibration data.
 - Investigated environmental and operational variability, missing data, anomaly detection, and data-driven structural health monitoring.
 - Identified VAE-OCSVM as the strongest-performing framework, achieving Precision = 0.996, Recall = 0.999, and Specificity = 0.970
- **Machine Learning-Based Prediction of Mechanical Properties of Steel-Fiber-Reinforced Concrete**
 - Developed and compared machine-learning and deep-learning models to predict the compressive and splitting tensile strengths of steel-fiber-reinforced concrete.
 - Conducted data collection, preprocessing, statistical analysis, model development, performance evaluation, and interpretation of results.

Education

M.Sc. in Civil-Structural Engineering

2019 - 2023

- Ferdowsi University of Mashhad (FUM), Mashhad, Iran
- **Thesis:** Machine Learning-Based Compressive Strength Modeling of Steel Fiber-Reinforced Concrete
- Supervisor: Prof. Mansoor Ghalehnovi

B.Sc. in Civil Engineering

2013 - 2019

- Sadjad University of Technology (SUT), Mashhad, Iran

Technical Skills

Engineering Software

- ABAQUS
- SAP2000
- ETABS
- SAFE

General Software

- Microsoft Word
- Microsoft Excel
- Microsoft PowerPoint

Programming

- Python
- MATLAB

Personal Skills

- Teamwork and collaboration
- Self-motivation
- Leadership
- Strong work ethic
- Project management
- Fast learning and adaptability

Language Skills

- **English:** Fluent
- **Persian:** Native

Personal Interests

- Travelling
- Reading
- Volleyball
- Movies

Selected Professional Certificates

- 2022 Learning Python, LinkedIn Learning, [Certificate](#)
- 2022 Machine Learning Specialization, Stanford University, [Certificate](#)
- 2021 Deep Learning, Stanford University, [Certificate](#)
- 2022 Understanding and Visualizing Data with Python, University of Michigan, [Certificate](#)
- 2022 Fitting Statistical Models to Data with Python, University of Michigan, [Certificate](#)
- 2021 Introduction to TensorFlow for AI, Machine Learning, and Deep Learning, Stanford University, [Certificate](#)
- 2023 Mechanics of Materials I: Fundamentals of Stress & Strain and Axial Loading, [Certificate](#)
- 2023 Mechanics of Materials II: Thin-Walled Pressure Vessels and Torsion, [Certificate](#)

References

Available on Request